



Guangdong Technion

Israel Institute of Technology

广东以色列理工学院

Homework 8 - Deadline December 27th

1. Find the exact value of the improper integral

$$\int_{1+\sqrt{2}}^{\infty} \frac{(\sqrt{2}-1)x^2 + 4\sqrt{2} + 2}{x^4 + 2x^2 - 8} dx.$$

2. a) Find the values of p for which the following integral converges and evaluate the integral in those cases:

$$\int_0^1 x^p \ln x \, dx.$$

b) For each positive integer n , compute

$$\int_0^1 \left[(\ln x)^n + \frac{1}{x^{(n-1)/n}} \right] dx.$$

3. a) Find the length of the curve $y = \arcsin(e^{-x})$ for $0 \leq x \leq \ln(\sqrt{2})$.

b) Find the length of the curve $y = 3\sqrt{x}$ between $(1, 3)$ and $(9, 9)$.

4. a) Find the area of the revolution surface, generated by the rotation of the curve $8x^2 = y^2 - y^4$ about the y -axis.

b) Find the area of the revolution surface, generated by the rotation of the curve $y = \cosh(x)$ about the x -axis on $[-2, 2]$.